

B-1383 Answer

The Classical Identity

Recall the identity:

$$F_{2n-1} = F_n^2 + F_{n-1}^2 \quad (1)$$

Answer

Substituting (1) into the given equation, the right-hand side becomes $(F_n^2 + F_{n-1}^2)^2$, so the constraint is

$$(pF_n^2 + qF_{n-1}^2) \left(\frac{F_n^2}{p} + \frac{F_{n-1}^2}{q} \right) = (F_n^2 + F_{n-1}^2)^2$$

By the **Cauchy-Schwarz inequality** (setting $\mathbf{u} = (\sqrt{p}F_n, F_{n-1}/\sqrt{q})$ and $\mathbf{v} = (F_n/\sqrt{p}, F_{n-1}/\sqrt{q})$):

$$(pF_n^2 + qF_{n-1}^2) \left(\frac{F_n^2}{p} + \frac{F_{n-1}^2}{q} \right) = |\mathbf{u}|^2 |\mathbf{v}|^2 \geq (\mathbf{u} \cdot \mathbf{v})^2 = (F_n^2 + F_{n-1}^2),$$

with equality if and only if $\mathbf{u} \parallel \mathbf{v}$, i.e.

$$\frac{\sqrt{p}F_n}{F_n/\sqrt{p}} = \frac{\sqrt{q}F_{n-1}}{F_{n-1}/\sqrt{q}} \implies p = q.$$

Since the constraint asserts equality, conclude $p = q$. With $p = q$, compute each side of the equation to prove. The left hand side:

$$\frac{F_n^2}{p^{F_n+1}} + \frac{F_{n-1}^2}{q^{F_{n-1}+1}} = \frac{F_n^2 + F_{n-1}^2}{p^{F_n+1}} = \frac{F_{2n-1}}{p^{F_n+1}}.$$

The right hand side:

$$\frac{F_{2n-1}^{F_n+2}}{(pF_n^2 + qF_{n-1}^2)^{F_n+1}} = \frac{F_{2n-1}^{F_n+2}}{(p(F_n^2 + F_{n-1}^2))^{F_n+1}} = \frac{F_{2n-1}^{F_n+2}}{p^{F_n+1} \cdot F_{2n-1}^{F_n+1}} = \frac{F_{2n-1}}{p^{F_n+1}}.$$

Therefore the two sides are equal.

□